Lesson plan · for the teacher

Fair by design

Optimization · Level 4 · Lesson 2

Time
40 min

Materials
Board and chalk; A 3 × 3 table of pupils' scores for three class jobs, one copy per group; Three job cards per group

Learning goals

- Score a plan by its total, its worst-off and its gap.
- Find the plan that makes the least happy person as happy as possible.
- Measure the price of fairness and agree a rule together.

Activity: Who carries the cost?

1. Warm-up (5 min): Ask: if a plan makes the class happiest in total, is it fair? Collect yes and no answers with reasons.
2. Teach (8 min): Draw the table: Sara rates the jobs board 9, plants 2, library 6; Tunde 7, 8, 4; Hoda 4, 5, 2. Explain the total, the lowest score (the worst-off) and the gap.
3. Solve (12 min): Half the groups find the plan with the biggest total (Sara board, Tunde plants, Hoda library: 19). The other half find the plan with the best lowest score (Sara library, Tunde board, Hoda plants: 18, nobody below 5). There are only six plans, so groups can try them all.
4. Compare (8 min): Put both plans on the board. The fair plan costs 1 point of the total. Ask Hoda's side and Sara's side how each plan feels.
5. Agree (7 min): The class votes on the rule to use for real class jobs next month, and you write it on the noticeboard.

Use real class jobs and let pupils give their own scores if they like; keep the table to three pupils and three jobs so every plan can be checked by hand.

Discussion

- When is a small price of fairness worth paying, and when is a large one?
- Who should choose the fairness rule for our school's duties?
- How can fairness work over several terms, not just one?

Exit ticket

- Write one fairness rule for a school decision, and say whether it is a floor, a goal or a limit on the gap.

Try it at school tomorrow
Give groups a small table of how much three pupils like three class jobs (1 to 9). One group finds the plan with the best total; another makes the least happy pupil as happy as possible. Compare: what did fairness cost, and which plan would the class choose?

Fair by design

1 **A rota has the best total, but one teacher gets the worst duty every week. What do you change?**

◯ Add a floor: every score at least a value, then the best total ◯ Take that teacher off the rota

◯ Nothing: the total is the best possible

2 **Three teachers rate three duties (gate, lunch hall, library) from 1 to 9: pref[teacher][duty]. This model finds the best total. Change it so it makes the worst-off teacher as well off as possible: add a variable for the worst score, say every teacher's score is at least that, and maximise it.**

```
from ortools.sat.python import cp_model
pref = [[9, 2, 6],
        [7, 8, 4],
        [4, 5, 2]]
m = cp_model.CpModel()
x = [[m.new_bool_var("") for d in pref]
     for t in pref]
for i in range(3):
    m.add_exactly_one(x[i])
    col = [row[i] for row in x]
    m.add_exactly_one(col)
m.maximize(sum(pref[t][d] * x[t][d]
               for t in range(3)
               for d in range(3)))
s = cp_model.CpSolver()
st = s.solve(m)
print(s.status_name(st),
      int(s.objective_value))
```

Write your program:

3 **Making the worst-off as well off as possible always lowers the total.**

◯ True ◯ False

4 **Two plans share the best floor of 5. Complete step 2: keep the floor as a limit, then make the total as big as possible.**

```
# step 2: keep the floor, then the total
m.add(worst _________ 5)
m. _________(total)
s = cp_model.CpSolver()
s. _________(m)
```

Word bank: `maximize` `solve` `print` `>=` `<=` `minimize`

5 **Which of these put fairness into a model? Choose all that apply.**

☐ A floor: every person's score at least 5 ☐ A goal: make the lowest score as high as possible

☐ A limit on the gap between the best and worst score ☐ Keeping the scores secret so nobody complains

☐ Listing the teachers in alphabetical order

✎ **Think of a school decision where the same people always seem to carry the cost. Which fairness rule would you propose, and who should agree it?**

Check your answers

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1 **Add a floor: every score at least a value, then the best total**

The best total can hide who carries the cost.

2

```
from ortools.sat.python import cp_model
pref = [[9, 2, 6],
        [7, 8, 4],
        [4, 5, 2]]
m = cp_model.CpModel()
x = [[m.new_bool_var("") for d in pref]
     for t in pref]
worst = m.new_int_var(1, 9, "worst")
for i in range(3):
    m.add_exactly_one(x[i])
    col = [row[i] for row in x]
    m.add_exactly_one(col)
    score = sum(pref[i][d] * x[i][d]
                for d in range(3))
    m.add(score >= worst)
m.maximize(worst)
s = cp_model.CpSolver()
st = s.solve(m)
print(s.status_name(st),
      int(s.objective_value))
```

Every teacher's score must be at least worst, so making worst as large as possible lifts the lowest score.

3 **False**

Sometimes one plan does both. In the Teachers to classes lab, the best total (39) also gives the best lowest score (6).

4

```
# step 2: keep the floor, then the total
m.add(worst >= 5)
m.maximize(total)
s = cp_model.CpSolver()
s.solve(m)
```

worst >= 5 keeps every teacher at 5 or more; maximize(total) then picks, among the fairest plans, the one that is best for the group; solve runs the search.

5 **A floor: every person's score at least 5; A goal: make the lowest score as high as possible; A limit on the gap between the best and worst score**

Floors and gap limits are constraints; lifting the lowest score is a goal.